Social Insurance and Occupational Mobility

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Outline

Motivation

- Social insurance provides a cushion for workers.
- Differs across countries, affecting labor market dynamics.
- How can we model this effect?

Research Question

Main Question

How does social insurance effect occpational experimentation?

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Hypothesis

Providing more social insurance allows for riskier occupational choices.

Model

- Build upon Roy (1951) model of occupational choice.
- Add interaction between earnings risk, social insurance, and occupational choice.

Human Capital

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- Innate
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    - discovered through work experience

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- Innate
    - occupation specific
    - discovered through work experience
- General
    - applicable to all occupations
    - will experience occupation-specific shocks to this human capital

Model Environment

Household:

- Lives for S periods
- endowed one unit of time each period, with no leisure value
- workers dislike risk
- rank levels of consumption c according to a utility function $u(c)$

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Labor Market:

- J occupations, $j = 1, ..., J$
- workers can only work in one occupation at a time, but can switch between periods
- receive wage w_j per unit of human capital

Value Functions

Value of staying in occupation j:

$$V_s(\Omega_s, z, \epsilon, j) = \{u(c) + \beta \int W_{s+1}(\Omega_{s+1}, z', \epsilon', j') dF(\epsilon')\},$$

$$s.t.$$

$$c = T(w_j e^{\theta_j} e^z e^\epsilon)$$

$$z' = z + \epsilon$$

$$\Omega_{s+1} = \Omega_s$$

Value Functions, cont.

Value of switching to occupation j':

$$H_s(\Omega_s, \theta_{j'}, z, \epsilon, j') = \{u(c) + \beta \int W_{s+1}(\Omega_{s+1}, z', \epsilon', j') dF(\epsilon')\},$$

$$s.t.$$

$$c = T(w_{j'} e^z e^{\theta'_j} e^{\epsilon'_j} e^{-c(s,\kappa)})$$

$$z' = z + \epsilon'$$

$$\Omega_{s+1} = \{\Omega_s, j', \theta_{j'}\}$$

Equilibrium

$\Psi_{j,s}(\Omega_s, z, \epsilon)$ is the distribution of workers in occupation j at age s with history Ω_s, general human capital z, and shock ϵ. Ψ is defined for all $\Omega_s, z, \epsilon \in \mathbb{R}$

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Mass of newborns:

$$\Psi_{j,0}(\Omega_0, z, \epsilon) = \frac{1}{S} f_j \forall j \in \{1, ..., J\}$$

where f_j is the fraction of newborns with initial occupation specific ability θ_j.

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For $s = 0, ..., S$:

$$\Psi_{j,s+1}(\Omega_{s+1}, z', \epsilon) = \sum_{j'} \Psi_{j',s}(\Omega_s, z, \epsilon) I_{j,s}(j', \omega_s, \epsilon, z) \forall j \in \{1, ..., J\}$$

Equilibrium cont.

Define aggregate mass of efficiency units in occupation j at age s as:

$$N_j = \frac{1}{S}\sum_{s\in S}\int e^{z}e^{\theta_{j'}}e^{\epsilon_{j'}} + \frac{1}{S}\sum_{s\in S}\sum_{j\neq j'}\int e^{-c(s,\kappa)}d\Psi_{j,s}(\Omega_{s-1},z,\epsilon)$$

Equilibrium cont.

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This let's us define the SCE consisting of

1. set of occupation level wages $\{w_j\}_{j=1}^{J}$
2. occupational populations $\{\Psi_j\}_{j=1}^{J}$
3. set of intermediate goods prices $\{p_j\}_{j=1}^{J}$
4. set of occupational labor inputs $\{N_j\}_{j=1}^{J}$
5. occupation-specific decision rules $\{I_{j,s}\}_{j=1,s=0}^{J,S}$
6. value functions $\{V_s\}_{s=0}^{S}$

Equilibrium Conditions

1. Above value functions solve optimization problems
2. Labor inputs N_j are the solution to the intermediate producer optimnization problem
3. Intermediate goods quantities X_j solve the final goods producer's problem
4. Prices p_j equate supply and demand of intermediate goods
5. Wage in occupation j is the marginal product of an efficiency unit in that occupation s.t.

$$w_j = \alpha_j N_j^{\alpha_j - 1} \Pi_{j' \neq j} \{ N_j'^{\alpha_j'} \}$$

6. Labor markets clear at occupational level
7. In occupation j, Ψ_j is the stationary distribution
8. Final goods market clears by Walras' Law

Data and Calibration

Data Used

- Cross-National Equivalent File (CNEF)
- Panel Study of Income Dynamics (PSID)
- German Socio-Economic Panel (GSOEP)

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Calibration requires variance of earnings shocks to be parameterized. This is done via estimating various moments

Estimation

- earnings (log): $y_{ijt} = \alpha_{ij} + \beta_j X_{ijt} + u_{ijt}$
- decomposition of risk, $u_{ijt} = \eta_{ijt-1} + \omega_{ijt}$
- ω_{ijt} is permanent and follows a random walk, η_{ijt-1} is a transitory shock which is i.i.d.
- tax function: $y_a = \phi_0 y_p^{1-\phi_1}$
- panel earnings: $y_{it} = \alpha_i +$
- probability of switching: $P_{it} \equiv Pr(y_{it} = 1 | v_{i\hat{\bar{t}}-1}) = \psi(v_{i\hat{\bar{t}}-1}; \beta)$
- mobility cost function: $c(s, \kappa) = \kappa_0 +_{1s} +\kappa_{2s}^2$

Baseline vs Counterfactual

TABLE 5
MODEL SUMMARY: BASELINE VERSUS COUNTERFACTUALS

	United States		Germany	
	Baseline	Taxes GER	Baseline	Taxes USA
Avg. occ. changes	3.968	4.293	0.452	0.345
Avg. mob. rate	0.187	0.199	0.026	0.020
Var. log earnings	0.703	0.719	0.179	0.178
Aggregate output	0.728	0.752	0.381	0.372
Relative to Baseline				
Avg. occ. changes (Δ)	0.324		−0.108	
Avg. mob. rate (Δ)	0.012		−0.006	
Var. log earnings (Δ)	0.016		−0.001	
Aggregate output (Δ%)	3.28%		−2.26%	

NOTE: The table presents the results of the quantitative model. It shows the value of the average number of occupational changes, the mobility rate, the variances of log earnings, and aggregate output for the United States (columns 2 and 3) and Germany (columns 4 and 5). The values of columns 2 and 4 refer to the baseline case. Columns 3 and 5 refer to the counterfactual exercise in which each country has the tax policy of the other: Taxes GER is the case of the United States with the tax code of Germany, and Taxes USA is the case of Germany with the tax code of the United States. The first panel presents the levels and the second the change with respect to the baseline case. Avg., average; mob., mobility; occ., occupational; Var., variance.

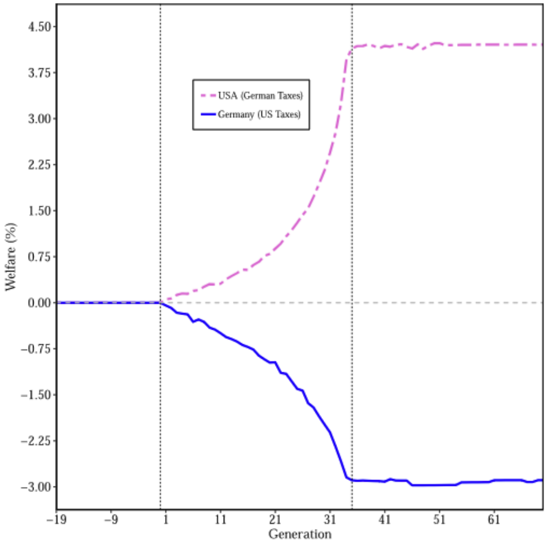
Baseline vs Counterfactual - Earning and Share by Occupation

TABLE 6
OCCUPATIONAL EARNINGS AND SHARES: BASELINE VRESUS COUNTERFACTUALS

	United States		Germany	
	Baseline	Taxes GER	Baseline	Taxes USA
Occ. shares				
Safe (S)	0.321	0.321	0.251	0.259
Medium (M)	0.460	0.455	0.418	0.407
Risky (R)	0.219	0.2224	0.332	0.334
Mean earnings				
Safe (S)	0.476	0.491	0.351	0.336
Medium (M)	0.674	0.708	0.367	0.366
Risky (R)	1.210	1.214	0.421	0.408

NOTE: The table presents the results of the quantitative model for the United States (columns 3 and 4) and Germany (columns 5 and 6). It shows the value of the occupational shares (first panel) and mean earnings (second panel) in each of the three occupations considered (the 12 occupations grouped in three groups according to their level of risk): safe (S), medium (M), and risky (R) groups. The values of columns 3 and 5 refer to the baseline case. Columns 4 and 6 refer to the counterfactual exercise in which each country has the tax policy of the other: Taxes GER is the case of the United States with the tax code of Germany, and Taxes USA is the case of Germany with the tax code of the United States. The first panel presents the levels and the second the change with respect to the baseline case. Occ, occupation.

Welfare Analysis



Main Results: US

- More social insurance leads to a mobility rate of 20%, a 1.3 pp. increase than the baseline
- Workers switch on average 4.3 times in their lifetimes, an 8.2% increase
- We see substitution towards higher risk occupations, leading to a higher average earnings
- Welfare rises by 4.15% under the German scheme via consumption smoothing and better occupational matches

Main Results: Germany

- Less social insurance leads to a decrease in average mobility rate from 2.6% to 2%
- Switching decreases from 0.45 to 0.35 times in a lifetime, due to US tax scheme discouraging occupational experimentation
- Mean earnings decrease in all occupations, most notably in the riskiest ones
- We see workers sort from medium risk to low risk occupations, leading to a polarity of risk types

Conclusion

- Uncertainty in earnings and social insurance are important factors in occupational choice
- Cross-country differences in social insurance can explain differences in occupational mobility and earnings
- Policy implications: designing social insurance programs to balance risk protection and incentives for occupational experimentation